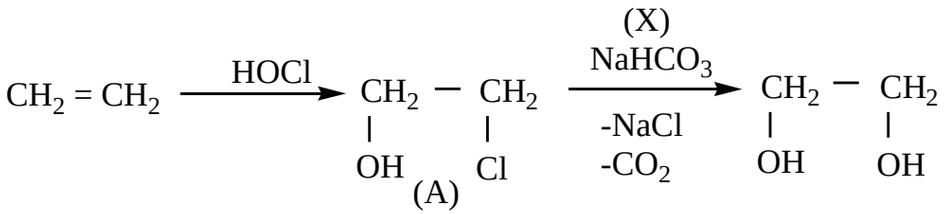
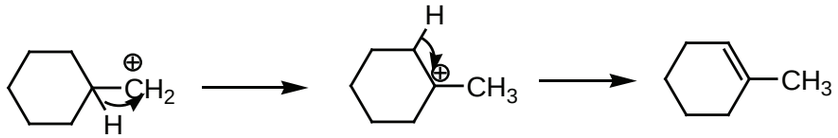
4.



5. (c) 4 – nitrophenol
6. Option (b) saytzeff rule

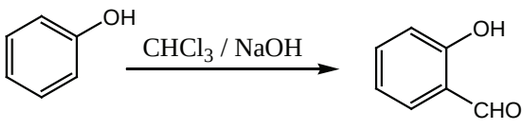


major product

7. Carbolic acid is

   a) phenol

8. Riemer – Tiemann reaction (option (c))



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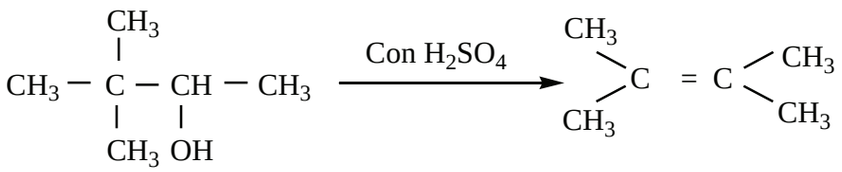
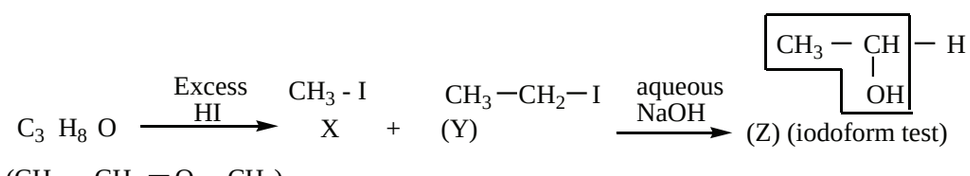


Option (b)

10. Option (a)
11. Option (a)
12. $CH_3 - CH_2 - OH \xrightarrow{PCl_5} CH_3 - CH_2 - Cl \xrightarrow{ale.KOH} CH_2 = CH_2 \xrightarrow{H_2SO_4/H_2O} CH_3 - CH_2 - OH$

    (Z) ethanol

13. Cyclic alcohol → sodium cyclic alkoxide → williamson ether synthesis option (c)
14. Option (d) phenol
15. Option (a)
16. Option (c)
17. Option (d)
18. Option (c)
19.



Option (d)

Answers.indd 315 11-01-2021 17:36:52